# Research References and Bibliography

## 1. Web Research References

### W1. Three common machine learning misconceptions. - GFI Software

• **Website:** gfi.ai

• **URL:** https://gfi.ai/three-common-machine-learning-misconceptions

• **Description:** Oct/13/2026

Eloquens AI now built Directly Into GFI KerioConnect

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Sep/25/2026

Eric Vaughan’s op-ed – Why tenure alone no longer predicts...

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Sep/22/2026

AI LIVE: The London Summit will bring together more than...

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Linkedin Profile [...] Conclusion

Machine learning is a rapidly evolving field with immense potential. By dispelling these common misconceptions, we can develop a more accurate understanding of what machine learning entails and how it can be effectively applied. As we continue to harness the power of machine learning, it is crucial to approach it with a critical and informed perspective.

Previous: State of adoption: Understanding the...Next: Demystifying AI: Separating Fact from...

### *RELATED*

Oct/13/2026 [...] One of the most prevalent misconceptions is that machine learning and artificial intelligence (AI) are interchangeable terms. While they are related, they are not the same.

Machine learning is a subset of AI that focuses on using algorithms to learn from data and make predictions. It involves training models on vast amounts of data, allowing them to identify patterns and make decisions without being explicitly programmed. Some examples of machine learning applications include:

• **Search relevance score:** 0.6864757

## W2. The 2026 Guide to Machine Learning | IBM

• **Website:** www.ibm.com

• **URL:** https://www.ibm.com/think/machine-learning

• **Description:** Machine learning (ML) is the subset of artificial intelligence (AI) focused on algorithms that can “learn” the patterns of training data and, subsequently, make accurate inferences about new data. This pattern recognition ability enables machine learning models to make decisions or predictions without explicit, hard-coded instructions. Machine learning, and in particular deep learning, is the backbone of most modern AI systems. [...] In this comprehensive guide, you will find a collection of machine learning-related content such as educational explainers, hands-on tutorials, podcast episodes and much more.

## Get started

Overview

As the first step in your journey, explore introductory machine learning explainers to obtain a high-level understanding.

Learn more

Data science for machine learning

Explore the fundamental data science and statistical principles that power machine learning use cases.

Learn more [...] # The 2026 Guide to Machine Learning

Published 22 September 2025

A neat stack of blue books arranged in a gradient pattern against a solid blue background. The minimalist design emphasizes symmetry and color uniformity. No visible text or numeric values are present.

By Cole Stryker, Fangfang Lee, Dave Bergmann and Mark Scapicchio

• **Search relevance score:** 0.64639384

## W3. Conceptual Framework of Common Misconceptions Regarding Generative AI in Elementary School Students Using Concurrent Think-Aloud Protocols

• **Website:** www.mdpi.com

• **URL:** https://www.mdpi.com/2073-431x/15/7/443

• **Description:** Skip to Content

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21 Pages

# Conceptual Framework of Common Misconceptions Regarding Generative AI in Elementary School Students Using Concurrent Think-Aloud Protocols

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Computers

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## Article Metrics

### *Article Access Statistics*

Journal Statistics

Multiple requests from the same IP address are counted as one view. [...] 8. European Commission: Directorate-General for Education, Youth, Sport and Culture. Ethical Guidelines on the Use of Artificial Intelligence (AI) and Data in Teaching and Learning for Educators; Publications Office of the European Union: Luxembourg, 2022; Available online: (accessed on 25 April 2026).

• **Search relevance score:** 0.51603514

## W4. Artificial intelligence applications in education: Natural language processing in detecting misconceptions | Education and Information Technologies | Springer Nature Link

• **Website:** link.springer.com

• **URL:** https://link.springer.com/article/10.1007/s10639-024-12919-1

• **Description:** Springer Nature

© 2026 Springer Nature [...] ")) and Long and Zeng (2022. Beginning deep learning with TensorFlow: Work with Keras. MNIST Data Sets, and Advanced Neural Networks.

")), ML comprises three types of learning algorithms: supervised, unsupervised, and reinforcement learning. [...] classified using Machine Learning (ML), one of the AI tools, and supervised learning algorithms. It was concluded that 175 teacher candidates did not have sufficient knowledge about the concept of greenhouse effect. It was found that the AI algorithm with the highest accuracy rate and used to predict teacher candidates' misconceptions was Multilayer Perceptron (MLP). Furthermore, through the Enhanced Ensemble Model Architecture developed by researchers, the combination of ML algorithms has

• **Search relevance score:** 0.459586

## W5. Machine Learning: Simply Explained

• **Website:** www.alexanderthamm.com

• **URL:** https://www.alexanderthamm.com/en/blog/machine-learning

• **Description:** semi-supervised learning, which uses a mixture of labeled and unlabeled data examples

self-supervised learning, which generates new labels from the data, and then learns how to apply those labels

## Common Machine Learning Algorithms

In machine learning, we use a wide range of algorithms, and different algorithms are well-suited to different tasks and types of data.

### *Linear Regression*

Linear regression is arguably one of the most common algorithms that we use in machine learning. [...] Common algorithms that fall under this ML approach are clustering (which helps us discover segments or groups in a dataset), as well as various dimension reduction techniques like principal component analysis (which helps simplify a dataset, often in preparation for other techniques like visualization or supervised learning).

### *Reinforcement Learning [...] Machine learning is a subfield of artificial intelligence and computer science that focuses on creating systems that learn and improve in the absence of explicit rules. So whereas traditional software is typically programmed to perform specific steps with explicit instructions (e.g., with loops, if/else statements, etc.), machine learning systems learn how to perform well and make decisions.*

• **Search relevance score:** 0.40457857

## W6. Shaping the Future of Learning: Education Readiness for ...

• **Website:** reports.weforum.org

• **URL:** https://reports.weforum.org/docs/WEF_Shaping_the_Future_of_Learning_2026.pdf

• **Description:** Global risks landscape. Adverse outcomes of AI technologies, 2026 F I G U R E 6 Shaping the Future of Learning: Education Readiness for the Age of AI 14 models hallucinate, they tend to use more confident language than when providing information, making them 34% more likely to use phrases like “definitely,” “certainly,” and “without doubt” when generating incorrect information.45 In educational settings, learners risk internalizing a cognitive norm in which plausibility replaces truth. When a [...] AI’s output, especially commercial tools not adapted for learning, might invert this relationship, instead encouraging acceptance rather than inquiry.46 These vulnerabilities are not uniform across age groups. Primary-school children are less likely to question an authoritative-sounding source and are more susceptible to forming lasting misconceptions.47 Older students, though more capable of evaluating AI output, may choose not to do so due to academic pressure.48 And educators are not exempt. [...] 44. Choi, A., & Mei, K.X. (2025, March 21). What are AI hallucinations? Why AIs sometimes make things up. The Conversation. 45. Basta, R. (2026, February 15). AI Hallucination Statistics: Research 2026. Suprmind.

46. Zhai, C., Wibowo, S., & Li, L.D. (2024). The effects of over-reliance on AI dialogue systems on student’s cognitive abilities: a systemic review. Smart Learning Environments, 11(3).

• **Search relevance score:** 0.39124367

## W7. Machine learning, explained | MIT Sloan

• **Website:** mitsloan.mit.edu

• **URL:** https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained

• **Description:** Supervised machine learning models are trained with labeled data sets, which allow the models to learn and grow more accurate over time. For example, an algorithm would be trained with pictures of dogs and other things, all labeled by humans, and the machine would learn ways to identify pictures of dogs on its own. Supervised machine learning is the most common type used today. [...] Bias and unintended outcomes

Machines are trained by humans, and human biases can be incorporated into algorithms — if biased information, or data that reflects existing inequities, is fed to a machine learning program, the program will learn to replicate it and perpetuate forms of discrimination. Chatbots trained on how people converse on Twitter can pick up on offensive and racist language, for example. [...] Madry pointed out another example in which a machine learning algorithm examining X-rays seemed to outperform physicians. But it turned out the algorithm was correlating results with the machines that took the image, not necessarily the image itself. Tuberculosis is more common in developing countries, which tend to have older machines. The machine learning program learned that if the X-ray was taken on an older machine, the patient was more likely to have tuberculosis. It completed the task,

• **Search relevance score:** 0.33073103

## W8. Designing the 2026 Classroom: Emerging Learning Trends in an AI-Powered Education System

• **Website:** www.facultyfocus.com

• **URL:** https://www.facultyfocus.com/articles/teaching-with-technology-articles/designing-the-2026-classroom-emerging-learning-trends-in-an-ai-powered-education-system

• **Description:** “Rooted in learning science, [Kyron] adapts to each learner’s responses, providing timely feedback, surfacing misconceptions early, and supporting diverse learning needs across environments,” Sheth explained.

## Opportunities and Challenges Students Face with AI [...] These institutions have understood that, in 2026, higher education will continue to adapt learning models to prepare students not only to use AI tools, but to thrive alongside them in the modern workforce. [...] “The future is educators empowered by AI, not overshadowed by it”,he added.

In 2026, learning how to use these learning tools will be crucial. Experts, in fact, have said that the best way for schools to manage AI risks is through teacher training and student literacy lessons.

By that measure, schools and districts are falling behind: although most teachers and students use AI, the CDT report shows that under half have received no training or guidance from their institutions.

• **Search relevance score:** 0.31053802

## W9. Artificial Intelligence and the Future of Teaching and ...

• **Website:** www.ed.gov

• **URL:** https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf

• **Description:** uses data to make a recommendation because it may appear to be more objective and authoritative, even if it is not. Although this perspective can be useful, it can be misleading. A human view of agency, pursuing goals, and reasoning includes our human abilities to make sense of multiple contexts. For example, a teacher may see three students each make the same mathematical error but recognize that one student has an Individualized Education Program to address vision issues, another [...] tools, shopping recommendations, essay-writing capabilities, and other familiar applications, AI may enhance educational services. Second, urgency and importance arise through awareness of system-level risks and anxiety about potential future risks. For example, students may become subject to greater surveillance. Some teachers worry that they may be replaced—to the contrary, the Department firmly rejects the idea that AI could replace teachers. Examples of discrimination from algorithmic bias [...] now. Today, for example, if an ITS specializes in feedback as a student practices, a human teacher could still be responsible for motivating student engagement and self-regulation along with other aspects of instruction. In other contemporary examples, the computer ITS might focus on problem solving practice, while teachers work with students in small groups. Further, students can be in the loop with AI, as is the case with “open learner models”—a type of AI-enabled system that provides

• **Search relevance score:** 0.18862306

## W10. AI vs. Machine Learning: How Do They Differ? | Google Cloud

• **Website:** cloud.google.com

• **URL:** https://cloud.google.com/learn/artificial-intelligence-vs-machine-learning

• **Description:** Machine learning is a subset of artificial intelligence that automatically enables a machine or system to learn and improve from experience. Instead of explicit programming, machine learning uses algorithms to analyze large amounts of data, learn from the insights, and then make informed decisions. [...] In the context of this example, the goal of using ML in the overall system is not to enable it to perform a task. For instance, you might train algorithms to analyze live transit and traffic data to forecast the volume and density of traffic flow. However, the scope is limited to identifying patterns, how accurate the prediction was, and learning from the data to maximize performance for that specific task.

### *Artificial intelligence*

### *Machine learning [...] Hyper-Personalization and Real-Time Optimization: In e-commerce and retail, ML algorithms analyze vast amounts of customer data to identify patterns and forecast demand, while the overarching AI system uses those predictions to provide personalized product recommendations and optimize supply chain processes in real-time.*

• **Search relevance score:** 0.1372434

# 2. Book References

## B1. Hello world

- **Author:** Hannah Fry
- **Publisher:** Norton & Company, Incorporated, W. W.
- **First published:** 2018
- **Open Library Work ID:** OL19761312W
- **Edition ID:** OL29791271M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL19761312W

## B2. Machine learning for hackers

- **Author:** Drew Conway
- **Publisher:** O'Reilly Media
- **First published:** 2012
- **Open Library Work ID:** OL16921250W
- **Edition ID:** OL25537172M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL16921250W

## B3. Out of control

- **Author:** Kevin Kelly
- **Publisher:** Perseus Books
- **First published:** 1994
- **Open Library Work ID:** OL1952087W
- **Edition ID:** OL24930969M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL1952087W

## B4. Machine Learning for Beginners

- **Author:** Jason Knox
- **Publisher:** Independently Published
- **First published:** 2019

- **Open Library Work ID:** OL26517436W
- **Edition ID:** OL41653125M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL26517436W

## B5. Mastering machine learning with R

- **Author:** Cory Lesmeister
- **Publisher:** Packt Publishing
- **First published:** 2015
- **Open Library Work ID:** OL26198197W
- **Edition ID:** OL35349589M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL26198197W

## B6. Leadership, Management and Data Science

- **Author:** Jerry Ramonyai
- **Publisher:** Independently Published
- **First published:** 2022
- **Open Library Work ID:** OL31075390W
- **Edition ID:** OL42688167M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL31075390W

## B7. Genetic programming

- **Author:** []
- **Publisher:** Morgan Kaufmann Publishers
- **First published:** 1997
- **Open Library Work ID:** OL16925931W
- **Edition ID:** OL8606478M
- **Language:** eng
- **Open Library link:** https://openlibrary.org/works/OL16925931W

## B8. Digital planet

- **Author:** Not available
- **Publisher:** Not available
- **First published:** Not available
- **Open Library Work ID:** OL15901395W
- **Edition ID:** OL24808461M
- **Language:** Not available
- **Open Library link:** https://openlibrary.org/works/OL15901395W

# 3. YouTube Video References

No structured YouTube references were available.